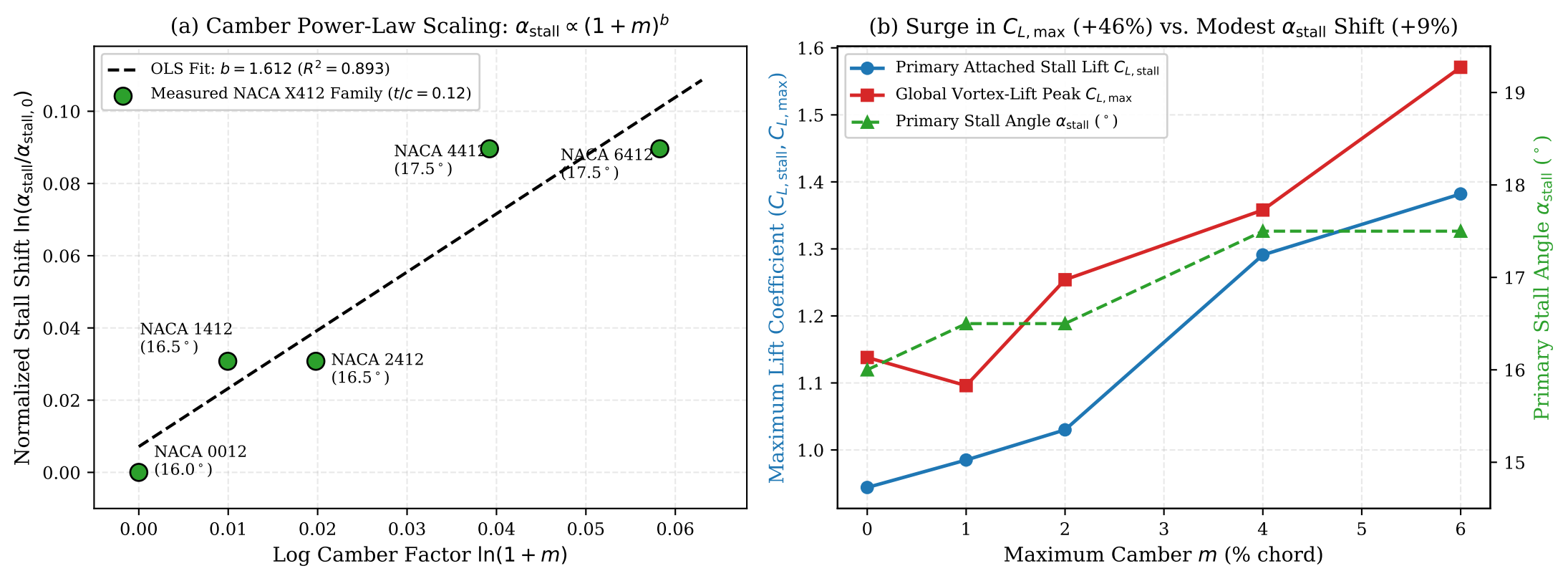
(a) Camber Power-Law Scaling: $\alpha_{stall} \propto (1+m)^b$
OLS Fit: $b = 1.612$ ($R^2 = 0.893$)
Measured NACA X412 Family ($t/c = 0.12$)
NACA 0012 (16.0°)
NACA 1412 (16.5°)
NACA 2412 (16.5°)
NACA 4412 (17.5°)
NACA 6412 (17.5°)
Normalized Stall Shift $\ln(\alpha_{stall}/\alpha_{stall,0})$
Log Camber Factor $\ln(1+m)$
0.00
0.02
0.04
0.06
0.08
0.10
0.00
0.01
0.02
0.03
0.04
0.05
0.06
(b) Surge in $C_{L,max}$ (+46%) vs. Modest α_{stall} Shift (+9%)
Primary Attached Stall Lift $C_{L,stall}$
Global Vortex-Lift Peak $C_{L,max}$
Primary Stall Angle α_{stall} (°)
Maximum Lift Coefficient ($C_{L,stall}$, $C_{L,max}$)
Primary Stall Angle α_{stall} (°)
Maximum Camber m (% chord)
1.0
1.1
1.2
1.3
1.4
1.5
1.6
15
16
17
18
19
0
1
2
3
4
5
6